Name: _____

WORKSHEET

Properties of the dot product

Consider the following vectors.

$\mathbf{a} = (3, 5)$	$\mathbf{b} = (-2, 1)$	$\mathbf{c} = (5, -4)$	$\mathbf{d} = (-4, 9)$	$\mathbf{e} = (-3, -5)$	$\mathbf{f} = (7, 3)$
$\mathbf{g} = 3\mathbf{i} - 2\mathbf{j}$	$\mathbf{h} = -4\mathbf{i} - 6\mathbf{j}$	$\mathbf{k} = -2\mathbf{i} + 4\mathbf{j}$	$\mathbf{l} = \mathbf{i} - 2\mathbf{j}$	$\mathbf{m} = -5\mathbf{i} - \mathbf{j}$	$\mathbf{n} = 4\mathbf{i} + 3\mathbf{j}$

1 Use the following pairs of vectors to demonstrate that the scalar product is commutative.

a **d** and **f**

b **c** and **a**

c **h** and **m**

d **n** and **g**

2 To prove that the scalar product is commutative ($\mathbf{p} \cdot \mathbf{q} = \mathbf{q} \cdot \mathbf{p}$):

a define general vectors **p** and **q**.

b find the LHS of the equation: $\mathbf{p} \cdot \mathbf{q}$

c find the RHS of the equation $\mathbf{q} \cdot \mathbf{p}$

d state the result.

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3 Use the following triplets of vectors to demonstrate that the scalar product is distributive over vector addition.

a **b**, **c** and **e**

b **a**, **f** and **d**

c **k**, **h** and **l**

d **m**, **g** and **n**

4 To prove that the scalar product is distributive over vector addition ($\mathbf{p} \cdot (\mathbf{q} + \mathbf{r}) = \mathbf{p} \cdot \mathbf{q} + \mathbf{p} \cdot \mathbf{r}$):

a define general vectors **p**, **q** and **r**.

b find the LHS of the equation: $\mathbf{p} \cdot (\mathbf{q} + \mathbf{r})$

c find the RHS of the equation $\mathbf{p} \cdot \mathbf{q} + \mathbf{p} \cdot \mathbf{r}$

d state the result.

5 Use the following pairs of vectors to demonstrate that $m(\mathbf{a} \cdot \mathbf{b}) = (m\mathbf{a}) \cdot \mathbf{b} = \mathbf{a} \cdot (m\mathbf{b})$.

a **c** and **f**, where $m = 5$

b **a** and **b**, where $m = 2$

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c **h** and **k**, where $m = -4$

d **f** and **g**, where $m = -3$

6 To prove that $m(\mathbf{p} \cdot \mathbf{q}) = (m\mathbf{p}) \cdot \mathbf{q} = \mathbf{p} \cdot (m\mathbf{q})$:

a Define general vectors **p** and **q**.

b find the LHS of the equation: $m(\mathbf{p} \cdot \mathbf{q})$

c find the middle section of the equation: $(m\mathbf{p}) \cdot \mathbf{q}$

d find the RHS of the equation: $\mathbf{p} \cdot (m\mathbf{q})$

e state the result.

7 Use the following vectors to demonstrate that $|\mathbf{a}| = \sqrt{\mathbf{a} \cdot \mathbf{a}}$.

a **d**

b **f**

c **n**

d **g**

8 To prove that $|\mathbf{p}| = \sqrt{\mathbf{p} \cdot \mathbf{p}}$:

a define a general vector $\mathbf{p}$

b find the LHS of the equation: $|\mathbf{p}|$

c find the RHS of the equation: $\sqrt{\mathbf{p} \cdot \mathbf{p}}$

d state the result.

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Answers

1 a $\mathbf{d} \cdot \mathbf{f} = \mathbf{f} \cdot \mathbf{d} = -1$

b $\mathbf{c} \cdot \mathbf{a} = \mathbf{a} \cdot \mathbf{c} = -5$

c $\mathbf{h} \cdot \mathbf{m} = \mathbf{m} \cdot \mathbf{h} = 26$

d $\mathbf{n} \cdot \mathbf{g} = \mathbf{g} \cdot \mathbf{n} = 6$

2 a $\mathbf{p} = (p_1, p_2)$ and $\mathbf{q} = (q_1, q_2)$

b $\mathbf{p} \cdot \mathbf{q} = p_1q_1 + p_2q_2$

c $\mathbf{q} \cdot \mathbf{p} = q_1p_1 + q_2p_2$

d Thus, $\mathbf{p} \cdot \mathbf{q} = \mathbf{q} \cdot \mathbf{p}$. **QED**

3 a $\mathbf{b} \cdot (\mathbf{c} + \mathbf{e}) = \mathbf{b} \cdot \mathbf{c} + \mathbf{b} \cdot \mathbf{e} = -13$

b $\mathbf{a} \cdot (\mathbf{f} + \mathbf{d}) = \mathbf{a} \cdot \mathbf{f} + \mathbf{a} \cdot \mathbf{d} = 69$

c $\mathbf{k} \cdot (\mathbf{h} + \mathbf{l}) = \mathbf{k} \cdot \mathbf{h} + \mathbf{k} \cdot \mathbf{l} = -26$

d $\mathbf{m} \cdot (\mathbf{g} + \mathbf{n}) = \mathbf{m} \cdot \mathbf{g} + \mathbf{m} \cdot \mathbf{n} = -36$

4 a $\mathbf{p} = (p_1, p_2)$, $\mathbf{q} = (q_1, q_2)$ and $\mathbf{r} = (r_1, r_2)$

b $\mathbf{p} \cdot (\mathbf{q} + \mathbf{r}) = (p_1, p_2) \cdot (q_1 + r_1, q_2 + r_2) = p_1(q_1 + r_1) + p_2(q_2 + r_2) = p_1q_1 + p_1r_1 + p_2q_2 + p_2r_2$

c $\mathbf{p} \cdot \mathbf{q} + \mathbf{p} \cdot \mathbf{r} = (p_1, p_2) \cdot (q_1, q_2) + (p_1, p_2) \cdot (r_1, r_2) = p_1q_1 + p_2q_2 + p_1r_1 + p_2r_2$

d Thus, $\mathbf{p} \cdot (\mathbf{q} + \mathbf{r}) = \mathbf{p} \cdot \mathbf{q} + \mathbf{p} \cdot \mathbf{r}$. **QED**

5 a $5(\mathbf{c} \cdot \mathbf{f}) = (5\mathbf{c}) \cdot \mathbf{f} = \mathbf{c} \cdot (5\mathbf{f}) = 115$

b $2(\mathbf{a} \cdot \mathbf{b}) = (2\mathbf{a}) \cdot \mathbf{b} = \mathbf{a} \cdot (2\mathbf{b}) = -2$

c $-4(\mathbf{h} \cdot \mathbf{k}) = (-4\mathbf{h}) \cdot \mathbf{k} = \mathbf{h} \cdot (-4\mathbf{k}) = 64$

d $-3(\mathbf{f} \cdot \mathbf{g}) = (-3\mathbf{f}) \cdot \mathbf{g} = \mathbf{f} \cdot (-3\mathbf{g}) = -45$

6 a $\mathbf{p} = (p_1, p_2)$ and $\mathbf{q} = (q_1, q_2)$

b $m(\mathbf{p} \cdot \mathbf{q}) = m(p_1q_1 + p_2q_2) = mp_1q_1 + mp_2q_2$

c $(m\mathbf{p}) \cdot \mathbf{q} = (mp_1, mp_2) \cdot (q_1, q_2) = mp_1q_1 + mp_2q_2$

d $\mathbf{p} \cdot (m\mathbf{q}) = (p_1, p_2) \cdot (mq_1, mq_2) = mp_1q_1 + mp_2q_2$

e Thus $m(\mathbf{p} \cdot \mathbf{q}) = (m\mathbf{p}) \cdot \mathbf{q} = \mathbf{p} \cdot (m\mathbf{q})$ **QED**

7 a $|\mathbf{d}| = \sqrt{\mathbf{d} \cdot \mathbf{d}} = \sqrt{97}$

b $|\mathbf{f}| = \sqrt{\mathbf{f} \cdot \mathbf{f}} = \sqrt{58}$

c $|\mathbf{n}| = \sqrt{\mathbf{n} \cdot \mathbf{n}} = 5$

d $|\mathbf{g}| = \sqrt{\mathbf{g} \cdot \mathbf{g}} = \sqrt{13}$

8 a $\mathbf{p} = (p_1, p_2)$

b $|\mathbf{p}| = \sqrt{(p_1)^2 + (p_2)^2}$

c $\sqrt{\mathbf{p} \cdot \mathbf{p}} = \sqrt{p_1 \times p_1 + p_2 \times p_2}$

d Thus $|\mathbf{p}| = \sqrt{\mathbf{p} \cdot \mathbf{p}}$ **QED**

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